Lower bounds for exceptional reductions

Pairs of elliptic curves and RM abelian surfaces

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Background: exceptional sets

Pairs of elliptic curves

Let $E, E'/K$ be non-isogenous. We study

$$S_{E,E'}(x) = \{v : Nv \leq x,\ E_v \sim E'_v\}.$$

- $E_v \sim E'_v$: geometrically isogenous reductions.
- **Charles '18:** $S_{E,E'}$ is infinite.
- Our goal: an explicit lower bound.

RM abelian surfaces

Let A/K be an abelian surface with real multiplication by F.

$$S_A(x) = \{v : Nv \leq x,\ A_v \text{ is nonsimple/split}\}.$$

- K number field.
- Split means A_v is geometrically isogenous to E_v^2.
- **Shankar–Tang '20:** S_A is infinite.
- Our goal: make the argument effective enough for recursion.

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Lang–Trotter philosophy

Heuristically: over $\mathbb{Q}$, $\{p \le x : \text{exceptional reduction at } p\} \asymp \frac{\sqrt{x}}{\log x}$.

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What Charles and Shankar–Tang proved

Charles: modular curve

- Work on $X(1) \times X(1)$.
- Intersect a fixed section $(j(E), j(E'))$ with Hecke correspondences t_N.
- Global height:

$$\widehat{\deg}\left(Z \cdot t_N Y\right) \asymp e_N \log N.$$

- Local terms are small for carefully chosen N.

Shankar–Tang: Hilbert modular surface

- Work on the Hilbert modular surface $\mathcal{H}$.
- Use compact Hirzebruch–Zagier divisors $T(r)$.
- Points on $T(r)$ carry extra special endomorphisms, hence give nonsimple reductions.
- Global height of $T_p[A]$ dominates local contributions.

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The Principle

global Arakelov growth $>$ old local contributions $\implies$ a new exceptional prime.

Results

⋆ **Work I: elliptic curves – from infinitude to** $\log^*$

Theorem: David–F.–Wang '26+

For non-isogenous elliptic curves $E, E'/K$,

$$|S_{E,E'}(x)| \gg_{E,E',K} \log^* x.$$

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⋆ **Work II: RM abelian surfaces – effective Shankar–Tang**

Theorem: F.–Wang–Xia, '26+

Let A/K be an RM abelian surface, viewed as a point $[A]$ on the associated Hilbert modular surface.

$$|S_A(x)| \gg_{A,K} \log^* x.$$

Proof Strategy

Effective Charles input.

- For old primes $A \subset S_{E,E'}$, construct many admissible $N \in [X^{1/2}, X]$.
- At every $v \in A$ and every archimedean place, the local contribution is controlled.
- Deep v-adic intersections produce many degree-N elements in a local Hom-lattice.

Local lattice threshold.

$$H_0(v) = \mathrm{Hom}_{\overline{\kappa(v)}}(E_v, E'_v), \qquad N_0(v) = \mu_{\mathrm{rk}\, H_0(v)}(H_0(v)).$$

Both ordinary and supersingular cases give explicit bounds for $N_0(p)$.

Recursive step

If $A = \{p_1, \ldots, p_k\}$ and

$$X > \max_{p_i \in A} \exp\big(C p_i^4 \log p_i \log\log p_i\big),$$

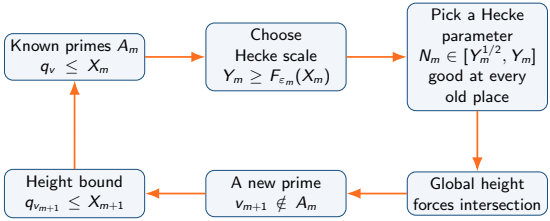
then every admissible $N \in [X^{1/2}, X]$ forces

$$p_{k+1} \in S_{E,E'} \setminus A.$$

A modular-polynomial height bound controls norm of the new place. We will iterates this one-step statement.

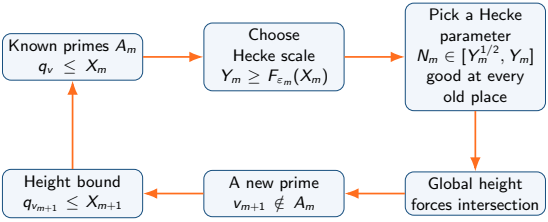
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The shared mechanism



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The shared mechanism



Why the output is $\log^*$

$F_\varepsilon(X)$ uniformly bounds the local cutoff for places with $q_v \le X$. The finite-height condition

$$\log^* F_\varepsilon(X) \le \log^* X + O_\varepsilon(1).$$

implies

$$\log^* F_{\varepsilon_m}(X_m) \le \log^* X_m + C, \qquad C \text{ independent of } m,$$

Thus after m steps we have produced $m \gg \log^* X_m$ exceptional primes, and

$$\#\{v : q_v \le x,\ v \text{ exceptional}\} \gg \log^* x.$$

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